

CHAPTER 2

Biological Classification

Introduction

- Five kingdom system of classification suggested by R.H. Whittaker is not based on (2014)
 - Complexity of body organisation
 - Presence or absence of a well defined nucleus
 - Mode of reproduction
 - Mode of nutrition

Kingdom Monera

- Which of the following is a correct statement? (2022)
 - Mycoplasma have DNA, Ribosome and cell wall
 - Cyanobacteria are a group of autotrophic organisms classified under Kingdom Monera
 - Bacteria are exclusively heterotrophic organisms
 - Slime moulds are saprophytic organisms classified under Kingdom Monera.
- Which of the following is incorrect about Cyanobacteria? (2020-Covid)
 - They lack heterocysts
 - They often form blooms in polluted water bodies
 - They have chlorophyll 'a' similar to green plants
 - They are photoautotrophs
- Oxygen is **not** produced during photosynthesis by (2018)
 - Nostoc*
 - Green sulphur bacteria
 - Cycas*
 - Chara*
- Which of the following are found in extreme saline conditions? (2017-Delhi)
 - Archaeobacteria
 - Eubacteria
 - Cyanobacteria
 - Mycobacteria
- Which among the following are the smallest living cells, known without a definite cell wall, pathogenic to plants as well as animals and can survive without oxygen? (2017-Delhi)
 - Bacillus*
 - Pseudomonas*
 - Mycoplasma
 - Nostoc*
- Methanogens belong to: (2016 - II)
 - Dinoflagellates
 - Slime moulds
 - Eubacteria
 - Archaeobacteria
- Which one of the following statements is wrong? (2016 - I)
 - Cyanobacteria* are also called blue-green algae
 - Golden algae are also called desmids

- Eubacteria are also called false bacteria
 - Phycomycetes are also called algal fungi
- The primitive prokaryotes responsible for the production of biogas from the dung of ruminant animals, include the: (2016 - I)
 - Halophiles
 - Thermoacidophiles
 - Methanogens
 - Eubacteria
 - True nucleus is absent in: (2015)
 - Vaucheria*
 - Volvox*
 - Anabaena*
 - Mucor*
 - The gut of cow and buffalo possess: (2015)
 - Methanogens*
 - Cyanobacteria*
 - Fucus*
 - Chlorella*
 - Cell wall is absent in: (2015 Re)
 - Funaria*
 - Mycoplasma*
 - Nostoc*
 - Aspergillus*
 - Archaeobacteria differ from Eubacteria in: (2014)
 - Mode of reproduction
 - Cell membrane structure
 - Mode of nutrition
 - Cell shape

Kingdom Protista

- Match List-I with List-II: (2024 Re)

List-I Organisms		List-II Mode of nutrition	
(A)	Euglenoid	(I)	Parasitic
(B)	Dinoflagellate	(II)	Saprophytic
(C)	Slime mould	(III)	Photosynthetic
(D)	<i>Plasmodium</i>	(IV)	Switching between photosynthetic and heterotrophic mode

Choose the **correct** answer from the options given below:

- A-III, B-IV, C-II, D-I
 - A-IV, B-II, C-I, D-III
 - A-IV, B-III, C-II, D-I
 - A-IV, B-II, C-III, D-I
- Select the **wrong** statement (2018)
 - Cell wall is present in members of Fungi and Plantae
 - Mushrooms belong to Basidiomycetes
 - Pseudopodia are locomotory and feeding structures in Sporozoans
 - Mitochondria are the powerhouse of the cell in all kingdoms except Monera

16. Which of the following organisms are known as chief producers in the oceans? (2018)
 a. Dinoflagellates b. Diatoms
 c. Cyanobacteria d. Euglenoids
17. Ciliates differ from all other protozoans in (2018)
 a. Using flagella for locomotion
 b. Having a contractile vacuole for removing excess water
 c. Using pseudopodia for capturing prey
 d. Having two types of nuclei
18. Select the wrong statement: (2016 - II)
 a. Diatoms are chief producers in the oceans
 b. Diatoms are microscopic and float passively in water
 c. The walls of diatoms are easily destructible
 d. 'Diatomaceous earth' is formed by the cell wall of diatoms.
19. Chrysophytes, Euglenoids, Dinoflagellates and Slime moulds are included in the kingdom: (2016 - I)
 a. Animalia b. Monera
 c. Protista d. Fungi
20. In which group of organisms the cell walls form two thin overlapping shells which fit together? (2015 Re)
 a. Euglenoids b. Dinoflagellates
 c. Slime moulds d. Chrysophytes

Kingdom Fungi

21. Which one of the following is not a criterion for classification of fungi? (2024)
 a. Mode of spore formation
 b. Fruiting body
 c. Morphology of mycelium
 d. Mode of nutrition
22. Match List-I with List-II. (2024)

List-I		List-II	
(A)	<i>Rhizopus</i>	(I)	Mushroom
(B)	<i>Ustilago</i>	(II)	Smut fungus
(C)	<i>Puccinia</i>	(III)	Bread mould
(D)	<i>Agaricus</i>	(IV)	Rust fungus

Choose the correct answer from the options given below:

- a. A-(III), B-(II), C-(I), D-(IV)
 b. A-(IV), B-(III), C-(II), D-(I)
 c. A-(III), B-(II), C-(IV), D-(I)
 d. A-(I), B-(III), C-(II), D-(IV)
23. Which of the following statements is correct? (2021)
 a. Fusion of protoplasts between two motile or non-motile gametes is called plasmogamy.
 b. Organisms that depend on living plants are called saprophytes.
 c. Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.
 d. Fusion of two cells is called Karyogamy.

24. Which of the following statements is **incorrect**? (2019)
 a. Morels and truffles are edible delicacies.
 b. *Claviceps* is a source of many alkaloids and LSD.
 c. Conidia are produced exogenously and ascospores endogenously.
 d. Yeasts have filamentous bodies with long thread-like hyphae.
25. After karyogamy followed by meiosis, spores are produced exogenously in (2018)
 a. *Neurospora* b. *Alternaria*
 c. *Agaricus* d. *Saccharomyces*
26. Which one of the following is wrong for fungi? (2016 - II)
 a. They are heterotrophic
 b. They are both unicellular and multicellular
 c. They are eukaryotic
 d. All fungi possess a purely cellulosic cell wall
27. One of the major components of cell wall of most fungi is: (2016 - I)
 a. Chitin b. Peptidoglycan
 c. Cellulose d. Hemicellulose
28. Which one of the following matches is correct? (2015)

a.	<i>Mucor</i>	Reproduction by Conjugation	Ascomycetes
b.	<i>Agaricus</i>	Parasitic fungus	Basidiomycetes
c.	<i>Phytophthora</i>	Aseptate mycelium	Basidiomycetes
d.	<i>Alternaria</i>	Sexual reproduction absent	Deuteromycetes

29. Choose the wrong statements: (2015 Re)
 a. *Neurospora* is used in the study of biochemical genetics
 b. Morels and truffles are poisonous mushrooms
 c. Yeast is unicellular and useful in fermentation
 d. *Penicillium* is multicellular and produces antibiotics
30. The imperfect fungi which are decomposer of litter and help in mineral cycling belong to: (2015 Re)
 a. Basidiomycetes b. Phycomycetes
 c. Ascomycetes d. Deuteromycetes
31. Which one of the following fungi contains hallucinogens? (2014)
 a. *Ustilago* sp. b. *Morchella esculenta*
 c. *Amanita muscaria* d. *Neurospora* sp.

Kingdom Animalia

32. Pick up the wrong statement: (2015 Re)
 a. Protista has photosynthetic and heterotrophic modes of nutrition
 b. Some fungi are edible
 c. Nuclear membrane is present in Monera
 d. Cell wall is absent in Animalia
33. Which one of the following living organisms completely lacks a cell wall? (2014)
 a. Blue-green algae b. *Cyanobacteria*
 c. Sea-fan (*Gorgonia*) d. *Saccharomyces*

Viruses, Viroids, Prions & Lichens

34. Mad cow disease in cattle and Cr Jacob disease in humans are due to infection by _____. (2022 Re)
 a. Prion b. Bacterium
 c. Virus d. Viroid
35. Which of the following is correct about viroids? (2020)
 a. They have free RNA without protein coat.
 b. They have DNA with protein coat.
 c. They have free DNA without protein coat.
 d. They have RNA with protein coat.
36. Which of the following statement is **incorrect**? (2019)
 a. Viroids lack a protein coat.
 b. Viruses are obligate parasites.
 c. Infective constituent in viruses is the protein coat.
 d. Prions consist of abnormally folded proteins.
37. Viroids differ from viruses in having: (2017-Delhi)
 a. DNA molecules with protein coat
 b. DNA molecules without protein coat
 c. RNA molecules with protein coat
 d. RNA molecules without protein coat
38. Which of the following statements is wrong for viroids? (2016 - I)
 a. They lack a protein coat
 b. They are smaller than viruses
 c. They causes infections
 d. Their RNA is of high molecular weight
39. Which of the following are most suitable indicators of SO₂ pollution in the environment? (2015 Re)
 a. Conifers b. Algae
 c. Fungi d. Lichens
40. Select the wrong statements: (2015 Re)
 a. W.M. Stanley showed that viruses could be crystallised
 b. The term '*Contagium vivum fluidum*' was coined by M.W. Beijerinck
 c. Mosaic disease in tobacco and AIDS in human being are caused by viruses
 d. The viroids were discovered by D.J. Ivanowsky
41. Viruses have: (2014)
 a. Both DNA and RNA
 b. DNA enclosed in a protein coat
 c. Prokaryotic nucleus
 d. Single chromosome
42. A location with luxuriant growth of lichens on the trees indicates that the: (2014)
 a. Location is not polluted b. Trees are very healthy
 c. Trees are heavily infested d. Location is highly polluted
43. Which of the following shows coiled RNA strand and capsomeres? (2014)
 a. Retrovirus b. Polio virus
 c. Tobacco mosaic virus d. Measles virus

Answer Key

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (a) | 4. (b) | 5. (a) | 6. (c) | 7. (d) | 8. (c) | 9. (c) | 10. (c) |
| 11. (a) | 12. (b) | 13. (b) | 14. (c) | 15. (c) | 16. (b) | 17. (d) | 18. (c) | 19. (c) | 20. (d) |
| 21. (d) | 22. (c) | 23. (a) | 24. (d) | 25. (c) | 26. (d) | 27. (a) | 28. (d) | 29. (b) | 30. (d) |
| 31. (c) | 32. (c) | 33. (c) | 34. (a) | 35. (a) | 36. (c) | 37. (d) | 38. (d) | 39. (d) | 40. (d) |
| 41. (b) | 42. (a) | 43. (c) | | | | | | | |